Root Canal Morphology of Permanent Premolars

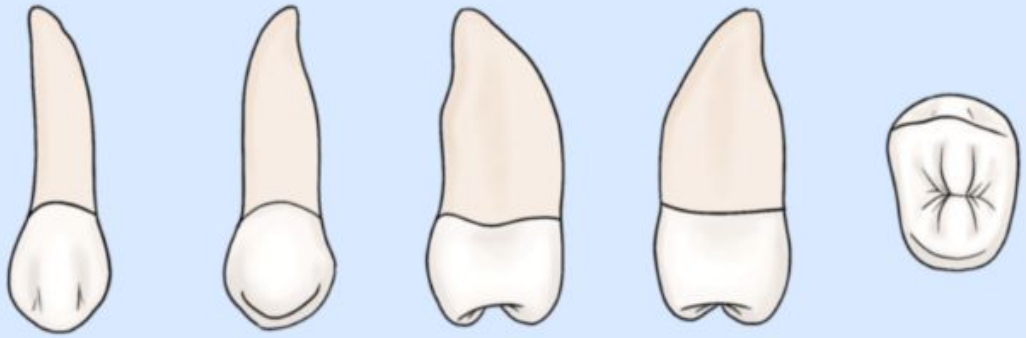
Learning Objectives:

- Describe the most common root and root canal anatomy of premolar teeth.
- For each tooth type, know the average length, number of roots, and most common root curvature directions.
- Characterize the most frequent variations in root and root canal anatomy of each tooth group.
- Identify the shape of the access opening
- Understand the cross sectional canal configuration

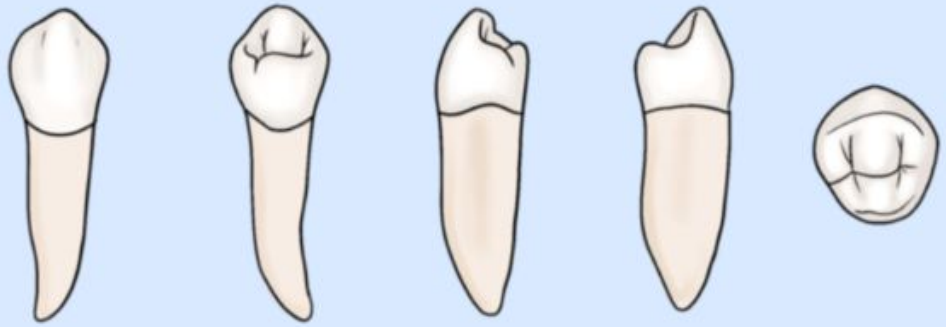
Maxillary 1st



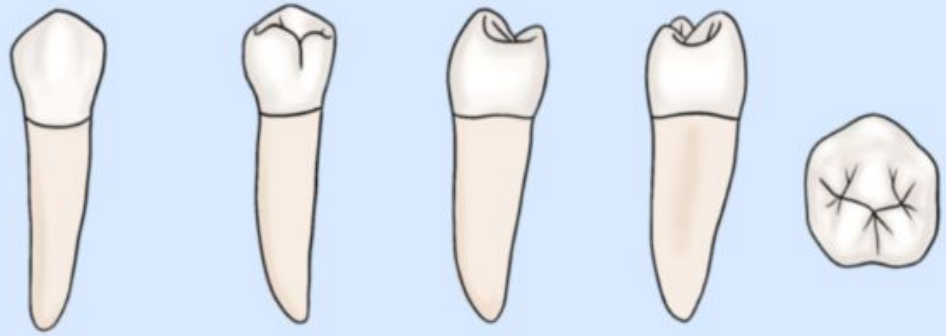
Maxillary 2nd



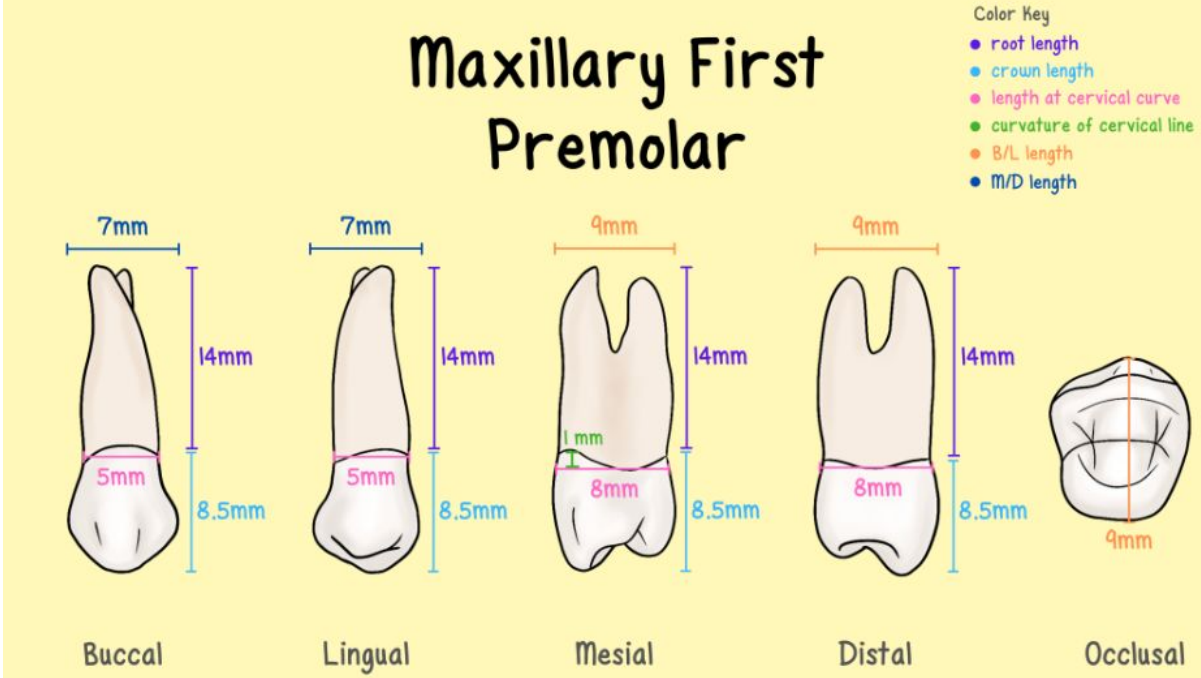
Mandibular 1st



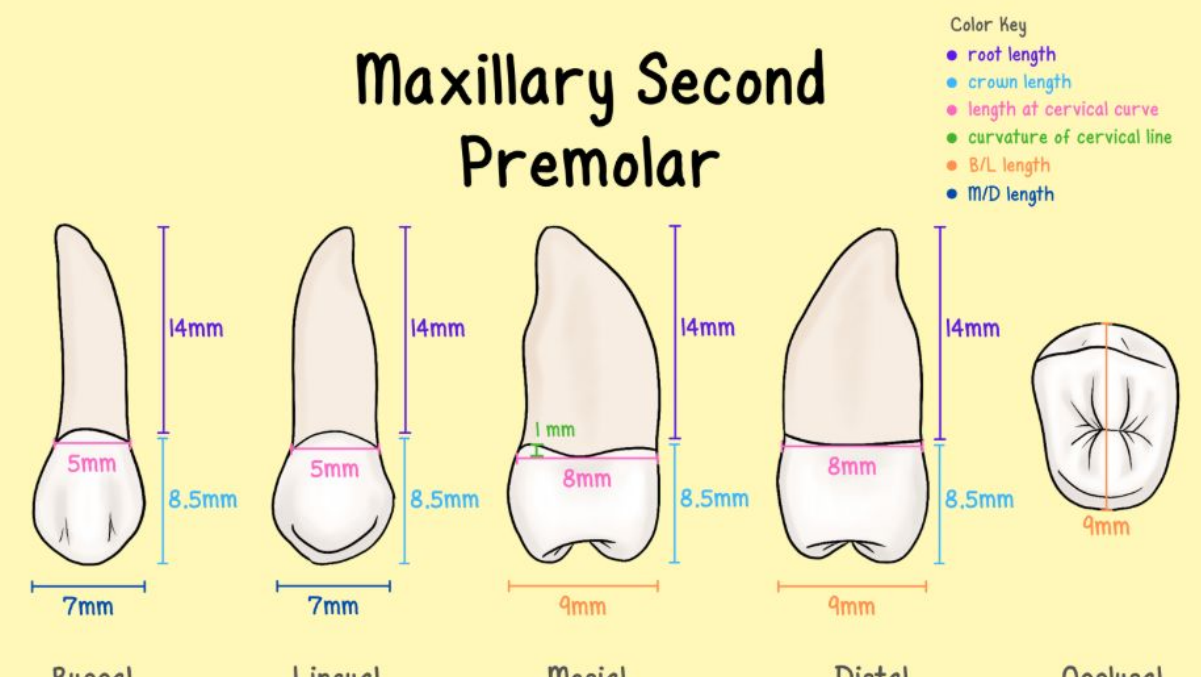
Mandibular 2nd



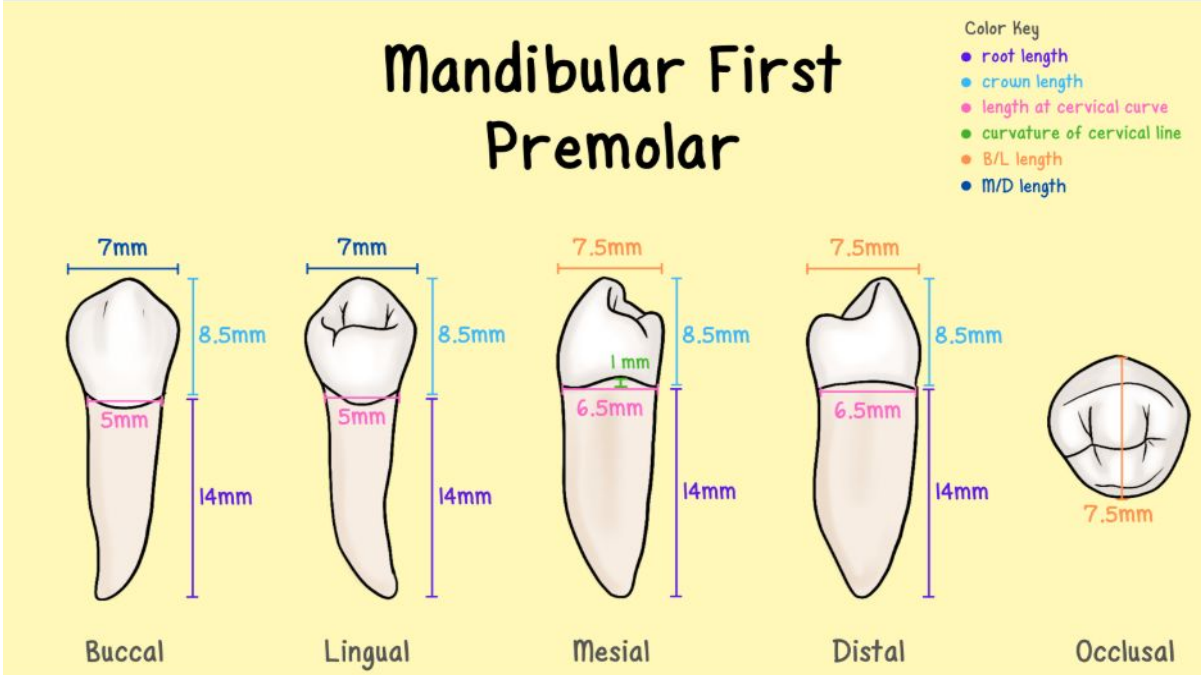
Maxillary First Premolar



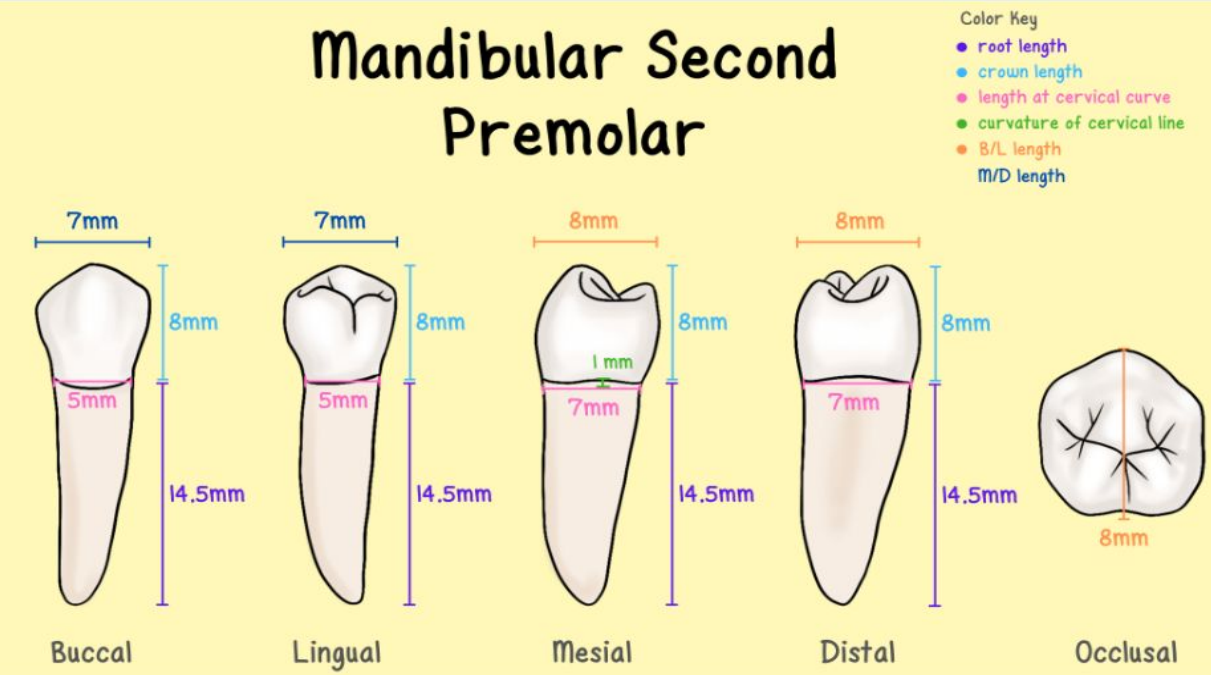
Maxillary Second Premolar



Mandibular First Premolar



Mandibular Second Premolar



I. Introduction to Premolar Endodontic Anatomy

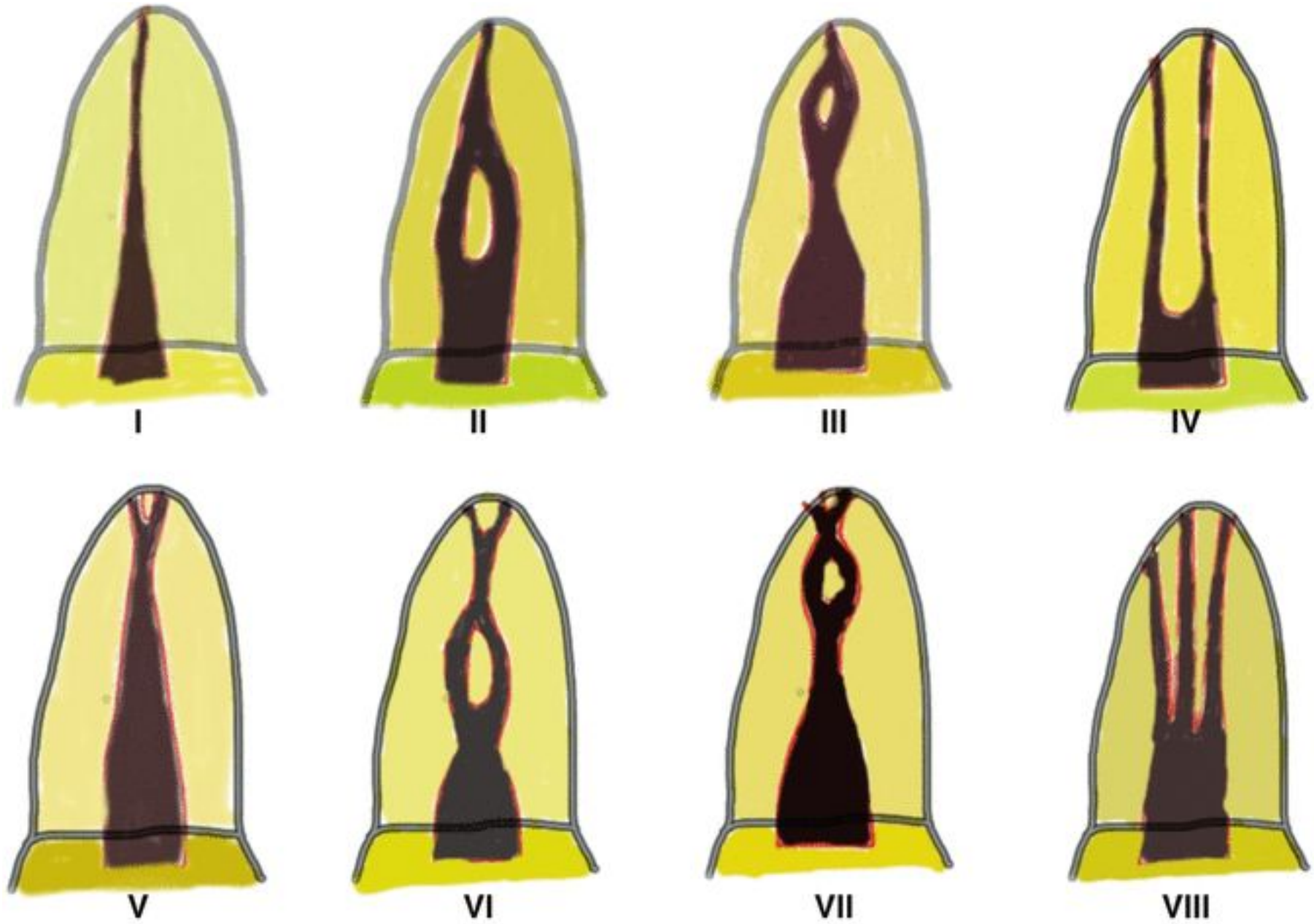
Clinical Significance:

Missed canals are a leading cause of endodontic failures.

Premolars and molars exhibit the highest incidence of aberrant internal anatomy across the human dentition.

Anatomical Mapping:

Proper instrumentation requires clinicians to anticipate complex root curvatures, multiple apical foramina, and variable canal configurations


I
II
III
IV
V
VI
VII
VIII

Classification Framework:

Internal configurations are standardized using Vertucci's Classification System (Types I to VIII):

Type I: 1 Single Canal → 1 Foramen (1)

Type II: 2 Separate Canals → Join at Apex into 1 Foramen (2→1)

Type III: 1 Canal → Splits into 2 → Rejoins into 1 Foramen (1→2→1)

Type IV: 2 Separate Canals → 2 Separate Foramina (2)

1. Maxillary First Premolar

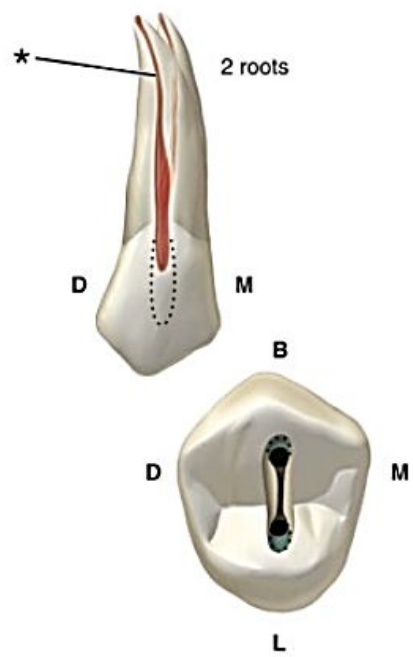
***Root Architecture*:** Predominantly **two roots** (Buccal and Palatal) connected by a prominent root trunk, though single-rooted variants exist.

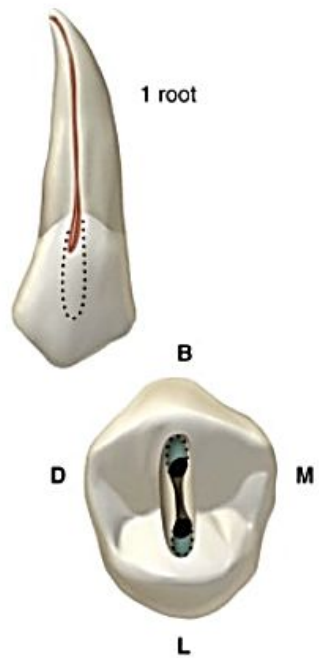
Internal Canal Configuration: Highly consistent **two-canal morphology** occurring in roughly 60–65% to up to 86.6% of cases.

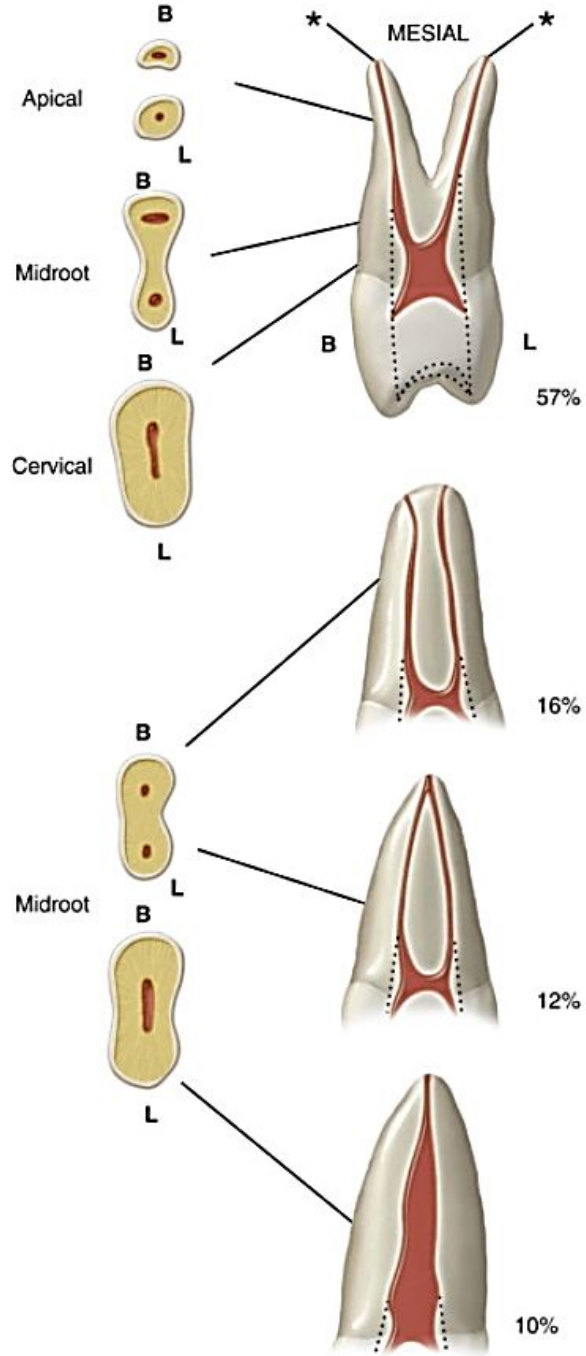
Dominant Classification: **Vertucci Type IV** (2-2 configuration) is the most prevalent layout.

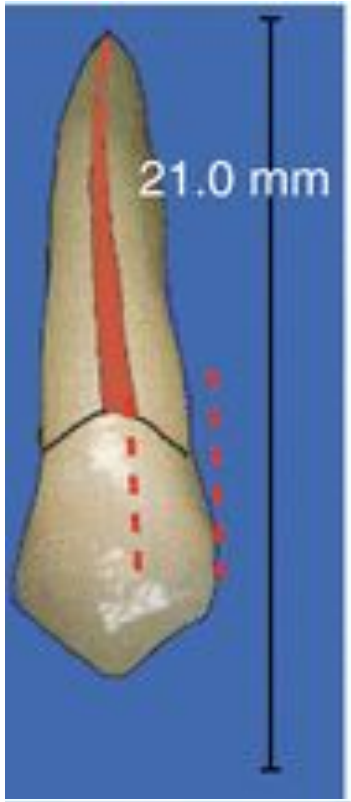
Pulp Chamber Features: Broadest in a buccopalatal dimension. A deep mesial developmental root concavity is present, increasing perforation risks during mechanical instrumentation.

Anatomical Aberrations: A rare three-rooted variant (incidence up to 6–9%) creates a "miniature molar" anatomy featuring separate mesiobuccal, distobuccal, and palatal canals.

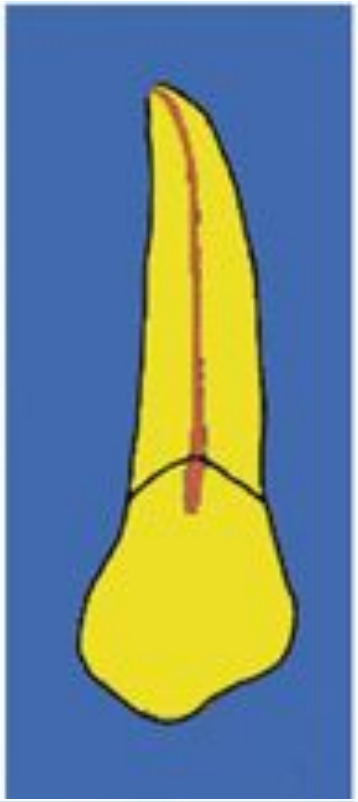

*
2 roots
D
M
B
D
M
L

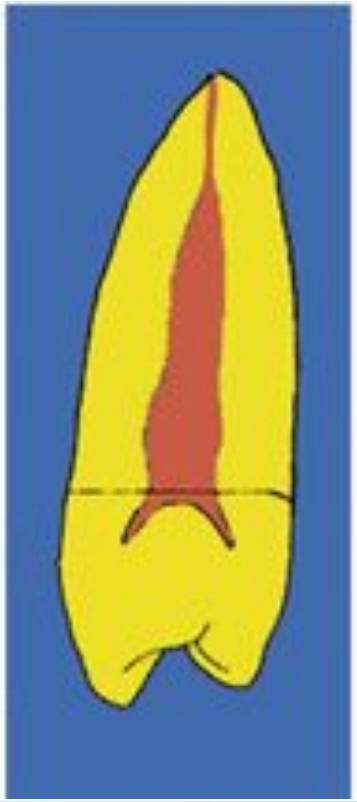

1 root
B
D
M
L

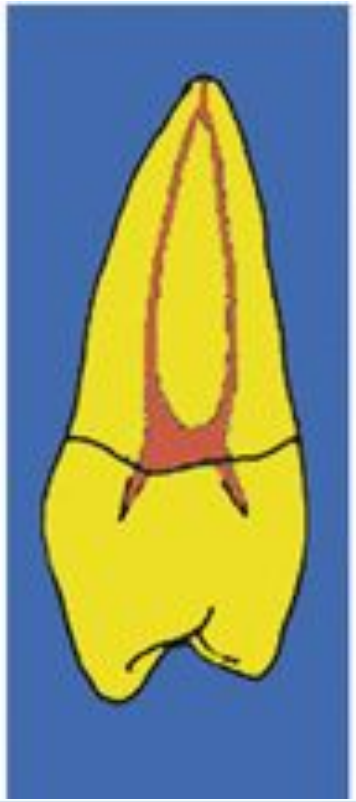

B
Apical
L
B
Midroot
L
B
Cervical
L
*
MESIAL
*
B
L
57%
16%
B
L
Midroot
B
L
12%
10%

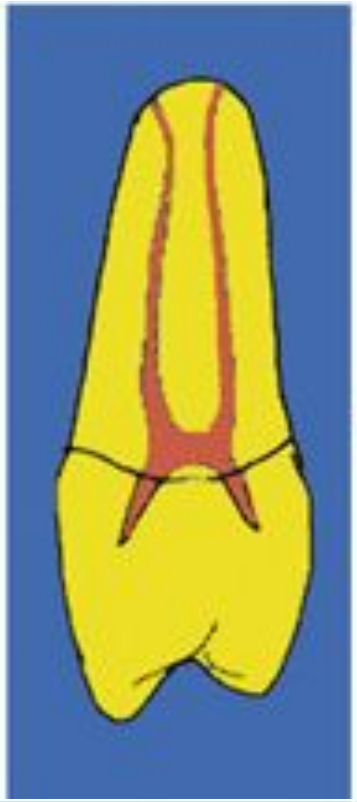








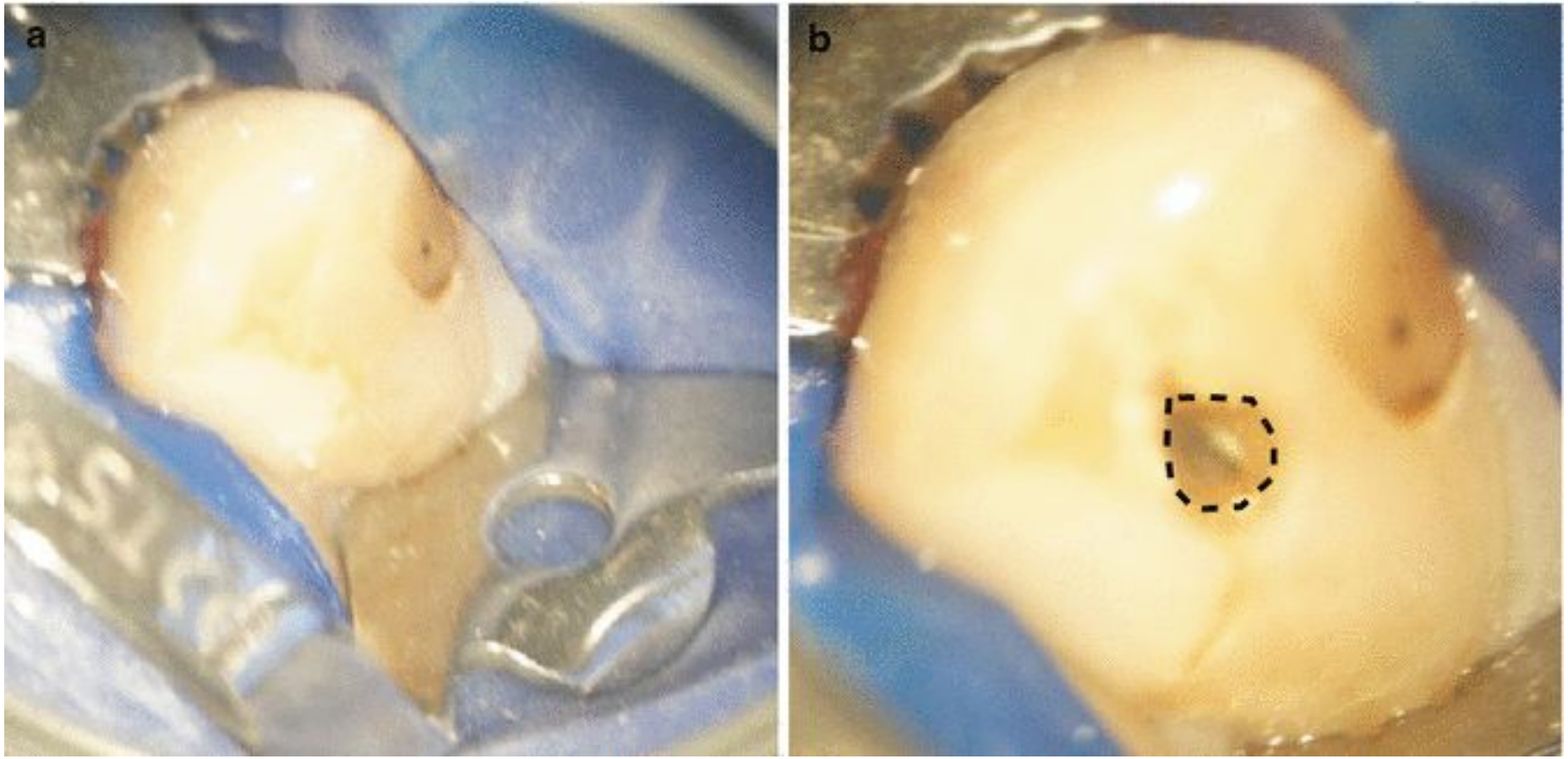








Number of canals/roots	Canal configuration and average length	Comments	Author(s)
Canals 1 26 % 2 65 % 3 3–6 % **Roots** 1 37 % 2 57 % 3 6 %	Vertucci type I,II or IV 21.0 mm	Access preparation oval in shape with greater extension towards buccal and lingual. Note that if three roots are present then buccal roots can be very thin. Anatomy of three rooted premolar similar to 1st molar in terms of canal positioning. Premolar teeth may have mesial concavity liable to strip perforation and post treatment fracture risk is high.	Vertucci


a
b

2. Maxillary Second Premolar

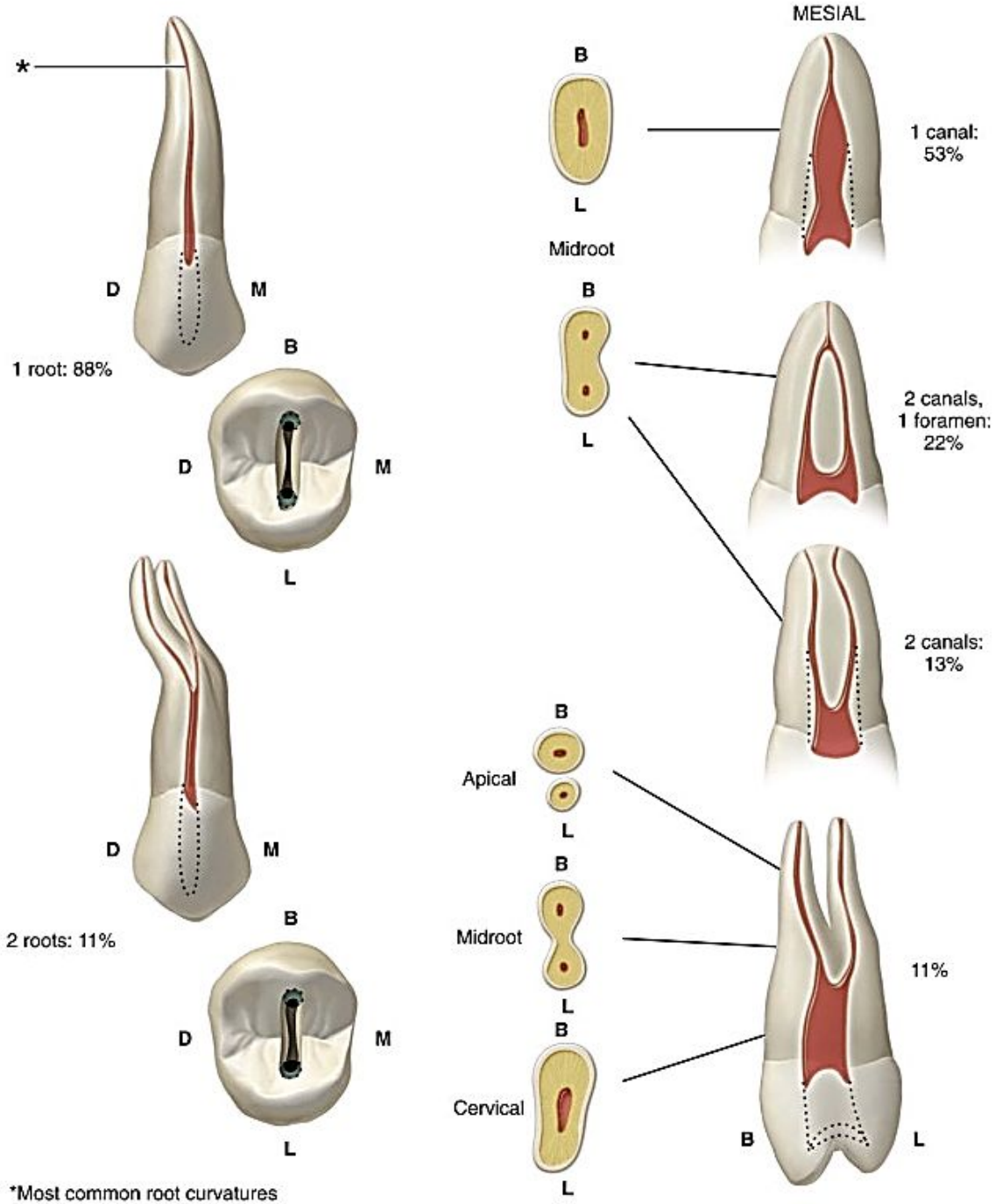
Root Architecture: Usually presents as a single root (roughly 66–77% of teeth)

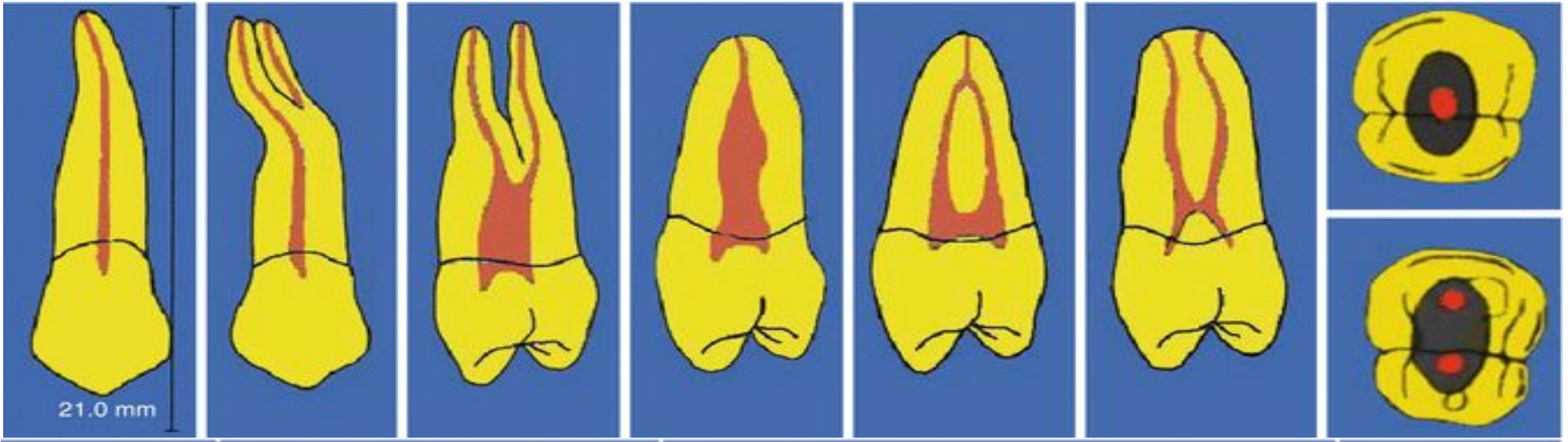
Internal Canal Configuration: Variable presentation; frequently presents with a **single canal (Type I)** or two canals that merge.

Dominant Classification: **Vertucci Type I** is the most common, followed closely by Type II and Type IV.

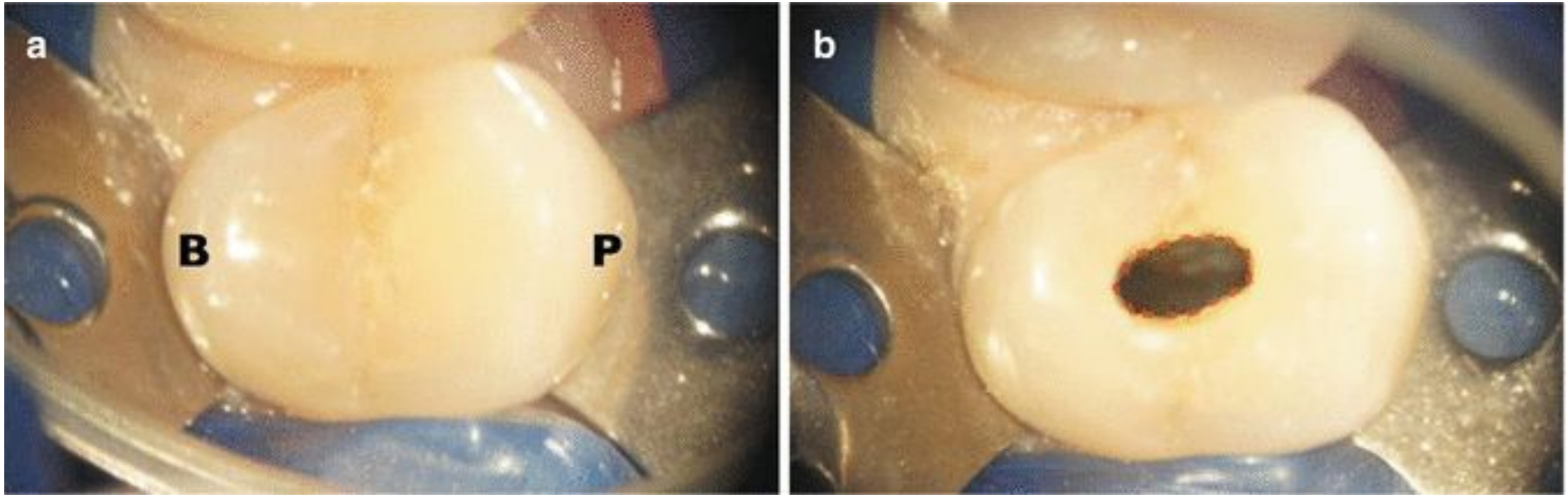
Pulp Chamber Features: The coronal pulp chamber is distinctly ribbon-shaped or elongated buccopalatally. It tapers cleanly down to a circular cross-section within the apical third.

Apical Splitting: In about 10–15% of specimens, a clinically single canal abruptly splits into two separate branches within the final 3–4 mm of the root tip.


*
D
M
1 root: 88%
B
D
M
L
D
M
2 roots: 11%
B
D
M
L
*Most common root curvatures
MESIAL
B
L
Midroot
1 canal:
53%
B
L
2 canals,
1 foramen:
22%
2 canals:
13%
B
Apical
L
B
Midroot
L
B
Cervical
L
11%
B
L



Number of canals/roots	Canal configuration and average length	Comments	Author(s)
Canals/ Foramina 1 1 48 % 2 1 27 % 2 2 24 % 3 1 1 % **Roots** 1 48 % 2 51 % 3 1 %	Vertucci type I,II or IV 21.0 mm	Access preparation oval in shape with greater extension towards buccal and lingual. Note that if three roots are present then buccal roots can be very thin. Anatomy of three rooted premolar similar to 1st molar in terms of canal positioning. Premolar teeth may have mesial concavity liable to strip perforation and post treatment fracture risk is high.	Vertucci Seelig & Gillis


a
B
P
b

1. Mandibular First Premolar

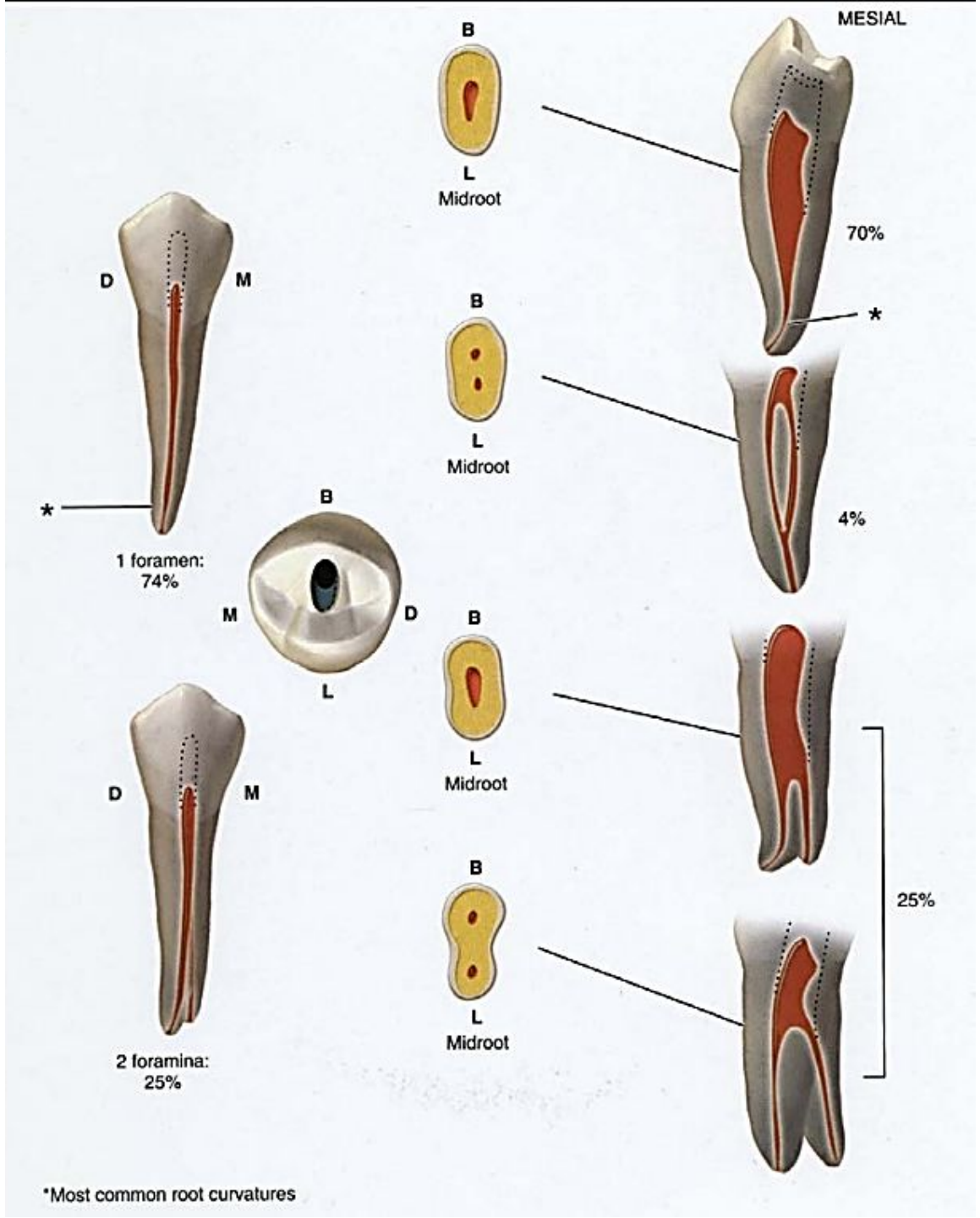
Root Architecture: Predominantly a single root (occurring in 77% to 98% of cases).

Internal Canal Configuration: Most frequently contains a single canal, but it possesses a high risk for sudden apical complexities.

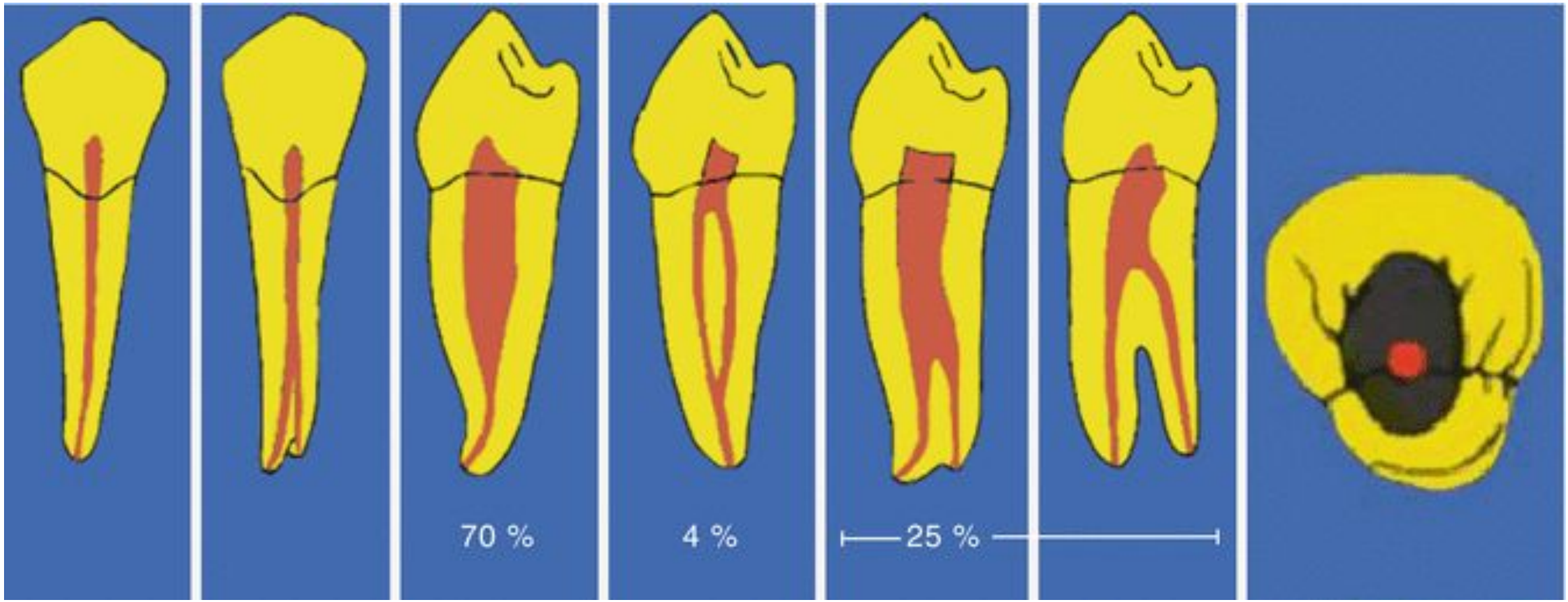
Dominant Classification: Vertucci Type I is standard (70–80%), but Type V (1→2 split) is a common alternative variant.

Pulp Chamber Features: The access cavity requires mesiodistal centering. The crown is significantly tipped lingually relative to the long axis of the root.

Anatomical Aberrations: Displays a distinct clinical tendency for C-shaped root canal complexes (with an incidence as high as 22.7% in specific populations).



*Most common root curvatures


70 %
4 %
25 %



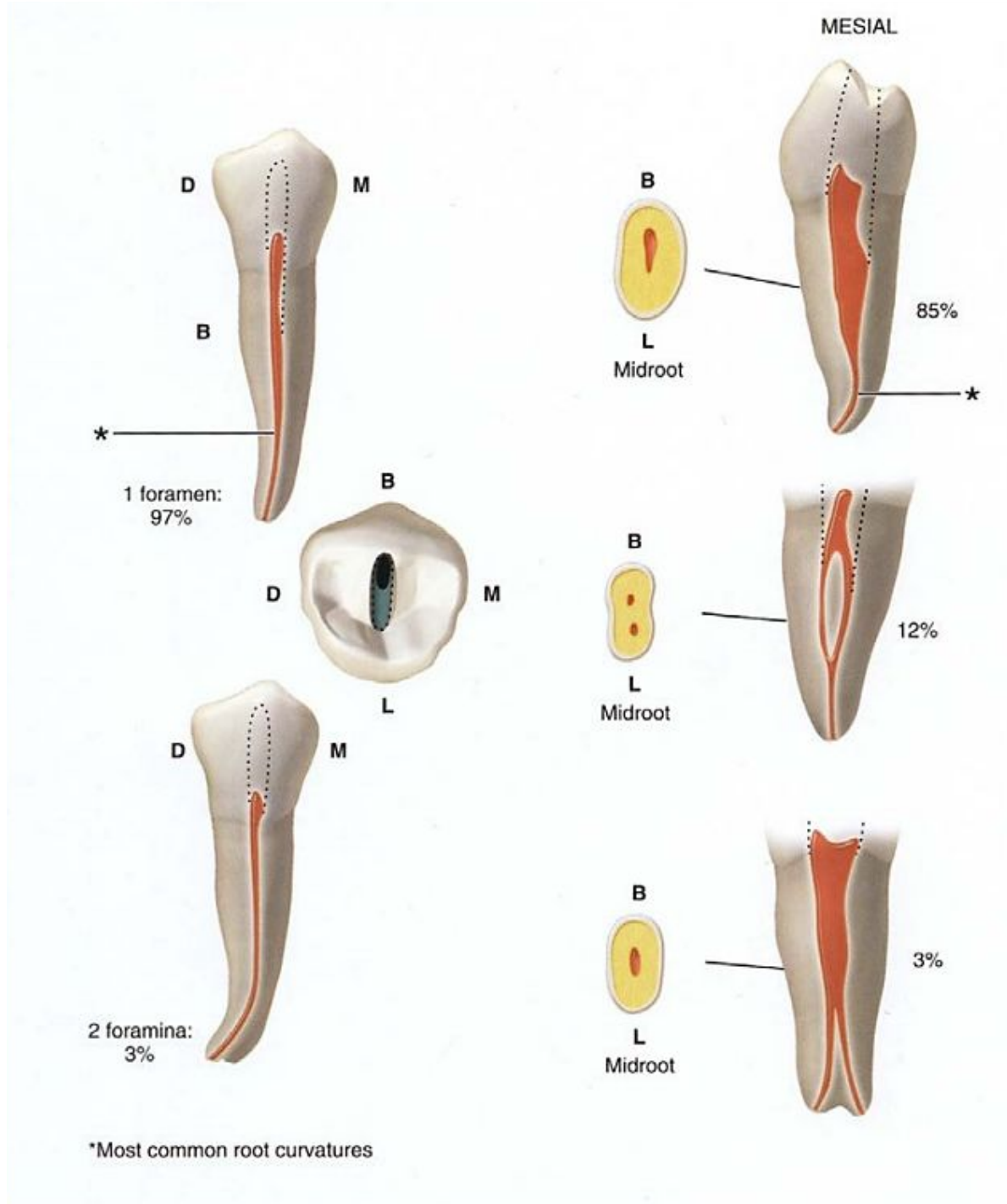
2. Mandibular Second Premolar

Root Architecture: Almost universally a single straight root (>93% of teeth).

Internal Canal Configuration: Highly uniform compared to its first premolar counterpart; it routinely hosts a single canal running to a single apical foramen.

Dominant Classification: Vertucci Type I represents over 78% to 93% of configurations.•

Pulp Chamber Features: Features a wider buccolingual dimension and a significantly more prominent lingual pulp horn than the first premolar. The direct boundary separating the pulp chamber floor and the root canal system is much cleaner and more readily distinguished.



*Most common root curvatures



IV. Summary Table

Tooth Type	Predominant Root Count	Most Common Vertucci Type	Common Anatomical Variations	Cross-Section Profile (CEJ)
Maxillary 1st Premolar	2 Roots (50–70%)	Type IV (2-2)	3 Roots (Molarization pattern)	Kidney-shaped / Disto-mesial Indentation
Maxillary 2nd Premolar	1 Root (75–90%)	Type I (1-1)	1 Root / 2 Canals (Type II & IV)	Oval
Mandibular 1st Premolar	1 Root (75–98%)	Type I (1-1)	C-shaped canals, severe Apical splits	Round / Oval / C-shaped
Mandibular 2nd Premolar	1 Root (90–97%)	Type I (1-1)	Rare 2 or 3 split canal variants	Broad, symmetrical Oval


thank you